MILITARY SPECIFICATION

SIGHTUNIT: M64A1

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of instrument known as Sightunit: M64A1 (9360168) used with US Army mortar systems, (see 6.1). Throughout this specification, this instrument is referenced as "sightunit". This sightunit is composed of Telescope, Elbow: 9360257 and Mount, Telescope: 9360169. This sightunit has radioactively illuminated scales, level vials, indices, and reticle.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwuise specified, the following specification and atandards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of
MIL-T-70521	-	Telescope, Elbow: 9360257
MIL-M-70453	-	Mount,Telescope: 9360169

FSC 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND (AMCCOM)

9360257	Telescope, Elbow: 9360257
9360168	Sightunit: 9360168
9360169	Mount, Telescope: 9360169
11750709	Adapter, Final Inspection, Azimuth

PACKAGING DATA SHEET

9360168	Packaging of Sightunit: M64A1

(Copies of drawings and packaging data sheets required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DSAM 4145.8	-	Radioactive Commodities in the DOD Supply System

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.)

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DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 CFR 171-190 - Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and other dangerous articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory).

Part 20: Standards for Protection against radiation.

Part 30: Rules of General applicability to Licensing of by-product material.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Radioactive material. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, possession, handling, and shipping of radioactive material.

3.2 First article. When specified, a sample shall be subjected to first article inspection (See 4.2 and 6.2).

3.3 Fabrication. The sightunit shall be manufactured in accordance with Drawing 9360168 and drawings pertaining thereto.

3.3.1 Telescope. The telescope shall be fabricated in accordance with Drawing 9360257 and drawings pertaining thereto, and shall meet the requirements of MIL-T-70521.

3.3.2 Mount. The telescope mount shall be fabricated in accordance with Drawing 9360169 and drawings pertaining thereto, and shall meet the requirements of MIL-M-70453.

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3.4 General specification. The contractor shall be responsible for adherence to, and compliance with, the requirements of MIL-F-13926 listed below:

a. Order of precedence.

b. Dimensions" and tolerances (see ANSI Y14.5).

c. Inorganic protective surface finishes.

d. Part identification and marking.

e. Workmanship.

3.5 Telescope clamping mechanism. When locked in place by applying a force of 10 to 12 in-lbs to the wingnut and 15 to 17 in-lbs to the locking lever, there shall be no rotation of the telescope within the inside diameter of the clamp when a 35 in-lb torque is applied. When locked in place with above forces, the clamping mechanism shall not rotate about its axis when a 15 in-lb torque is applied, When unlocked, the telescope and clamping mechanism shall rotate freely. This requirement shall be met at an ambient temperature between +60°F and +90°F (+16°C to +32°c) .

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three sightunits plus three each of all items covered by QAP. The sample shall be manufactured in the

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same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Quality conformance inspections.

4.3.1 Submission of product. Unless otherwise specified, by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Mount. The (see 3.3.2) shall have been inspected and accepted in accordance with the inspection provisions contained in MIL-M-70453 and the applicable Quality Assurance Provisions (QAP's) listed in the Technical Data Package (TDP).

4.3.2.2 Telescope. The telescope (see 3.3.1) shall have been inspected and accepted in accordance with the inspection provisions contained in MIL-T-70521 and the applicable QAP's listed in the TDP.

4.3.2.3 Sightunit. Examination and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in Tables I and II herein. Examination and tests for packaging, packing, and marking shall be in accordance with DSAM 4145.8 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I. 100 Percent (100%) inspection

No.	Characteristic	Requirement	Test procedure
1.	Telescope clamping mechanism	3.5	4.6.1

Note: The test in Table I shall be conducted at a temperature between +60° and +90°F (+16°C to +32°C).

4.3.3 Noncompliance. The right is reserved to reject any unit of product found defective. Rejected units may be repaired or corrected and resubmitted for inspection with the approval of, and in the manner specified by the procuring activity.

4.4 Quality conformance inspection sampling.

4.4.1 General. Three sightunits shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table II.

TABLE II

No.	Characteristic	Requirement	Test procedure
301.	Fabrication	3.3, 3.3.1 & 3.3.2	Applicable drawings and specifications - Visual
302.	General specification	3.4	MIL-F-13926 - Visual
303.	Telescope clamping mechanism	3.5	4.6.1

4.4.2 Noncompliance of sample. The right is reserved to reject any unit of product found defective. Should any one item of a sampling fail to meet the specified test requirements, acceptance of the represented inspection sample shall be suspended by the Government. Rejected samples shall be screened for all defective characteristics. Rejected units may be repaired or corrected and resubmitted for inspection with the approval of, and in the manner specified by the procuring activity.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities" reqquirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. The contractor shall submit for approval all inspection equipment designs in accordance with the terms of the contract (see 6.3).

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Telescope clamping mechanism. The Adapter, Final Inspection, Azimuth 11750709, together with a suitable stand, shall be utilized in performing the telescope clamping mechanism test. The stand shall have the necessary stability for positioning the adapter so that it maintains the mounting dovetail surface and dovetail axis of the mount in a vertical plane within plus or minus one degree.

4.5.2.3.2 Torque requirements. The torque wrench type tester for testing the telescope clamping mechanism shall have a range of zero to fifty inch-pounds. The accuracy of the torque wrench type tester shall be five percent throughout its entire zone of excursion. The torque wrench shall be equipped with suitable adapters to accommodate testing of the appropriate assembly.

4.5.2.3.3 Wall target. The wall target shall contain a vertical and horizontal crossline. It shall be positioned 6 to feet in front of the sightunit under test.

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4.5.2.3.4 Fabrication, setup and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instructions shall be prepared and submitted to the procuring agency.

4.6 Test methods and procedures.

4.6.1 Telescope clamping mechanism. Conduct this test with the mount positioned on the testing equipment specified in 4.5.2.3.1. Position the target so that the intersecting point of the vertical and horizontal target lines are coincident to the geometric axis of the clamp for-the telescope. Install the telescope in the clamping mechanism of the mount and lock the clamp by applying a force of 15 to 17 inch-pounds to the locking lever and 10 to 12 inch-pounds to the wingnut. By means of the elevation and azimuth knobs, adjust the line of sight of the telescope so that its vertical and horizontal reticle lines are superimposed on the image at the vertical and horizontal target lines. The telescope clamping mechanism shall conform to the requirements of 3.5 when the applicable torques are applied to the telescope body and the clamp for the telescope. Any rotation of the telescope within the clamp, or rotation of the clamping mechanism about its own axis shall be determined by the permanent displacement (lack of coincidence) of the telescope reticle lines with respect to the target image lines.

5. PACKAGING

5.1 Preparation of radioactive material. Handling, packaging, packing, shipping, and marking shall be in accordance with Packaging Data Sheet 9360168 and DSAM 4145.8. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Sightunit, M64A1 is used for laying the US Army mortar systems in deflection and elevation for direct and indirect fire modes.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation.

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c. Applicable stock number.

d. Applicable packaging data sheet number (see 5.1).

e. Provisions for First Article Testing.

f. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-10054. See 6.3.1.

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection shall be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armanent Research and Development Center, ATTN: AMSMC-QAF-I (D), Dover, NJ 07801-5001. When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of-transmittal:

a. The contract number.

b. The contract item (name, model number, etc.).

c. The designs remaining to be submitted and the expected date of submittal.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-70454	SIGHTUNIT: M64A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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